BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)
SATHYAMANGALAM - 638 401

Approved Questions
S2 Semester

22GE003 – BASICS OF ELECTRICAL ENGINEERING

Q. No.	Questions
	Unit 1
1.	In an electronics laboratory, students are performing experiments using glass rods, silk cloth, and ebonite rods to study the behavior of electric charges. During the experiment, they observe that when two charged bodies are brought close to each other, they either attract or repel depending on the type of charge present on them. The students also notice that the charge produced during rubbing remains on the object and influences nearby charged particles. To understand these observations, they analyze the fundamental properties that govern electric charge. Examine the fundamental properties of electric charge that explain attraction and repulsion between charged bodies.
2.	While working on electrostatic experiments, engineers observe that small plastic particles become charged when rubbed against a dry cloth. These charged particles then interact with other nearby particles and sometimes move or stick to surfaces. Engineers attempt to understand the characteristics of electric charge that allow it to influence other objects and produce observable effects in electrostatic systems. Assess the main characteristics that define electric charge in physical systems.
3.	In a physics demonstration, a charged balloon is brought near small pieces of paper placed on a table. The paper pieces immediately move toward the balloon and stick to its surface. Students observe that the balloon retains its charge for some time and continues to attract other lightweight objects. This leads them to investigate the properties of electric charge responsible for such interactions. Analyze how the properties of electric charge explain electrostatic attraction observed in this experiment.
4.	During a classroom experiment, two plastic rods are rubbed with wool and then brought close to each other. Instead of attracting, the rods move away from each other when placed nearby. Relate this observation to the property of electric charge.
5.	In a school physics laboratory, students perform an experiment using a plastic comb and small pieces of paper placed on a table. The comb is first rubbed against dry hair and then brought close to the paper pieces without touching them. The students observe that the paper pieces immediately jump toward the comb and stick to its surface. Even after some pieces fall back to the table, the comb continues to attract other nearby paper fragments. The teacher explains that this behavior occurs because the comb acquires electric charge through friction and produces a force that acts on nearby objects. Establish the role of electric charge in producing forces between objects.

6.	While studying electrostatics, students observe that some materials become positively charged while others become negatively charged after friction. Identify the basic classification of electric charges.
7.	During a classroom demonstration on electrostatics, a teacher rubs a plastic scale with a dry cloth and then slowly brings it close to tiny bits of thermocol and paper placed on a desk. Students observe that even without touching the pieces, the scale attracts them from a small distance and the particles move toward it. Some pieces briefly stick to the scale before falling off and being attracted again. This repeated interaction helps students understand that a charged body can influence nearby objects through electrostatic forces. Determine the physical property responsible for this electrostatic interaction.
8.	In an electrostatics experiment, students charge several small metal spheres by rubbing them with different materials. After charging, they bring the spheres into contact with each other and measure the resulting charge using an electrometer. They observe that the final charge on the combined system equals the sum of the individual charges originally present on each sphere. This observation leads them to analyze the principle that electric charges combine algebraically when multiple charged bodies interact. Examine the principle of additivity of electric charges and explain how total charge is determined when several charged bodies interact.
9.	During a laboratory investigation of electric charge, scientists observe that whenever an object becomes charged, the charge always appears in fixed discrete amounts rather than any arbitrary value. Even when very small charges are measured, they always appear as multiples of a fundamental unit. This observation leads researchers to study the concept that electric charge exists in discrete packets rather than continuous values. Assess the concept of quantization of electric charge and its significance in electrostatics.
10.	In an experimental setup, researchers charge three small metal spheres separately and later connect them through conducting wires. When the charges redistribute among the spheres, the final total charge measured in the system remains equal to the sum of the original charges present on the individual spheres. The scientists use this observation to verify a fundamental property of electric charge that governs how charges combine in a system. Analyze how the additivity property of electric charge explains the total charge of a system containing multiple charged bodies.
11.	During a physics demonstration, a teacher charges two metal spheres separately and later brings them into contact. Students observe that the charges combine and redistribute between the spheres. Relate this observation to the concept of additivity of electric charge.
12.	In a laboratory experiment using an electrometer, students observe that whenever an object gains or loses electric charge, the measured charge always appears as multiples of a specific smallest value. Even when the measured charge becomes very small, it never appears as a completely arbitrary number. This observation leads them to understand that electric charge is not continuous but exists in fixed discrete units related to a fundamental particle. Establish the concept that explains why electric charge appears only in fixed discrete amounts.
13.	While analyzing electrostatic measurements, a student notices that the charge measured on an object always appears as multiples of a basic unit. Identify the principle that explains this observation.

14.	During an electrostatics experiment, several charged particles are brought together in a closed system. Even though the charges redistribute among the particles, students notice that the total charge obtained by adding all individual charges remains equal to the combined charge of the system before interaction. This helps them understand how electric charges combine mathematically when multiple charged bodies are present in a system. Determine the principle that explains how the total charge of a system is obtained from individual charges.
15.	In a physics laboratory, students perform an experiment using two metal spheres placed on insulating stands. One sphere is positively charged while the other is neutral. When the two spheres are connected with a conducting wire, charge flows between them and eventually distributes evenly. After disconnecting the wire, the students measure the charges on both spheres and notice that although the distribution of charge has changed, the total amount of charge present in the system remains the same as before the interaction. This observation leads them to examine a fundamental law governing electric charge in physical systems. Examine the principle of conservation of electric charge and explain how it applies to the redistribution of charge in a system.
16.	During an electrostatic experiment, researchers place several charged particles inside a closed conducting container. When the particles interact with each other, electrons move between the particles and the container walls. Even though the charges redistribute in different parts of the system, measurements confirm that the overall charge inside the container remains unchanged throughout the process. Scientists use this observation to verify a fundamental law governing electric phenomena. Assess the significance of the conservation of electric charge in electrostatic systems.
17.	In particle physics experiments, scientists observe interactions between electrons and protons inside controlled environments. During collisions or interactions, particles may exchange charge or produce other particles, but careful measurements show that the total electric charge before and after the interaction always remains constant. This experimental evidence supports a fundamental conservation principle in physics. Analyze how particle interactions demonstrate the conservation of electric charge.
18.	Two metal plates initially carry different amounts of electric charge. When they are connected with a conducting wire, charge flows between them until equilibrium is reached. Relate this observation to the conservation of electric charge.
19.	In an electrostatic demonstration, a teacher charges a metal sphere and then brings it into contact with another neutral sphere. After contact, the charge spreads between the two spheres. When students measure the charges afterward, they find that the combined charge of the two spheres equals the original charge that was present on the first sphere. Even though the charge distribution has changed, the total charge remains unchanged. Establish the law that explains why the total charge remains constant in this interaction.
20.	During a laboratory experiment, students observe that when electrons move from one conductor to another, the charge on each conductor changes but the overall charge of the system remains unchanged. Identify the law governing this behavior.

21.	In a classroom demonstration, a teacher shows that when several charged bodies interact and exchange electrons, the charges on individual objects may change. However, when all the charges are added together after the interaction, the total value remains exactly the same as before the interaction occurred. This observation helps students understand a fundamental rule governing electric charge in nature. Determine the physical law illustrated in this situation.
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29.	In an electrical laboratory, engineers study the behavior of electric charge inside metallic conductors used in circuits. When a potential difference is applied across the ends of a copper wire, electrons inside the conductor begin to move slowly in a particular direction. Although individual electrons move randomly due to thermal motion, the applied electric field causes a net movement of electrons toward the positive terminal. This steady movement of electrons produces an electric current in the conductor. Engineers analyze how charges behave in conductors and how the drift of electrons contributes to current flow. Examine how electric charge moves in conductors and explain the concept of electron drift in current-carrying conductors.
30.	During the design of electrical wiring systems in buildings, engineers select materials such as copper and aluminum for conducting electricity. These materials allow electric charges to move easily through them when a voltage is applied. Engineers study the internal structure of these materials to understand why electrons can move freely and how this property allows the conductor to carry electric current efficiently. Assess the behavior of electric charges in conductors and explain why conductors allow the flow of electric current.
31.	In an electronics experiment, a long metallic wire is connected to a battery and a small bulb. When the circuit is completed, the bulb immediately glows, indicating that electric current is flowing through the wire. Scientists explain that inside the wire, millions of electrons move slowly under the influence of the electric field produced by the battery. This gradual movement of electrons in a preferred direction is responsible for current flow. Analyze the role of electron drift in the conduction of electric current in metallic conductors.
32.	In a classroom experiment, a copper wire is connected to a battery and a switch. When the switch is closed, electrons inside the wire begin moving under the influence of the electric field. Relate this observation to the concept of electron drift.
33.	During an electrical engineering practical session, students connect a metallic wire between the terminals of a battery and observe that the circuit allows electric current to pass easily. The instructor explains that in metallic conductors, a large number of free electrons are present that are not tightly bound to atoms. When an external electric field is applied, these electrons begin moving gradually in one direction through the material, allowing charge to flow continuously from one end of the conductor to the other. Establish why metallic conductors allow the flow of electric current.
34.	A metal wire connected to a battery carries electric current due to the motion of electrons inside the conductor. Identify the phenomenon responsible for this motion.
35.	During a physics demonstration on electrical conduction, a teacher explains that even though electrons inside a metal move randomly due to thermal motion, applying a voltage across the conductor causes a slight net movement of these electrons toward one direction. This gradual and directed movement of electrons throughout the conductor results in the steady flow of electric current in electrical circuits used in everyday devices. Determine the concept that explains the directed motion of electrons in a conductor under an electric field.

36.	During a thunderstorm, meteorologists observe the formation of large thunderclouds in the atmosphere. Inside these clouds, strong air currents cause water droplets, ice particles, and dust to collide frequently. These collisions result in the transfer of electric charges between particles, leading to the accumulation of positive charges in one region of the cloud and negative charges in another. As the difference in charge grows larger, strong electric fields develop between the cloud and the ground or between different parts of the cloud. Eventually, the accumulated charges discharge suddenly in the form of lightning. Examine how electric charges are formed and distributed within thunderclouds during a storm.
37.	Atmospheric scientists studying thunderstorms use weather balloons and sensors to monitor electrical activity inside clouds. Their observations show that powerful air currents move charged particles upward and downward within the cloud. As these particles collide and separate, large quantities of electric charge accumulate. When the electric field becomes sufficiently strong, a rapid discharge occurs either within the cloud or between the cloud and the Earth's surface. Assess the process by which charge separation in clouds leads to lightning formation.
38.	In meteorology research centers, scientists investigate the electrical nature of thunderclouds. They observe that ice crystals and water droplets move at different speeds within clouds due to air currents. These movements lead to repeated collisions between particles, causing charge transfer and separation within the cloud. As the cloud grows larger, the accumulation of charges increases the electric potential difference between the cloud and nearby objects such as tall buildings or trees. Analyze how particle collisions inside clouds contribute to the formation of atmospheric electric charges.
39.	During a thunderstorm, people often notice flashes of lightning followed by thunder. Scientists explain that these flashes are related to the presence of electric charges inside clouds. Relate lightning formation to the presence of electric charges in clouds.
40.	Meteorologists studying storm clouds observe that strong winds inside the clouds continuously move water droplets, ice crystals, and dust particles. These particles collide with each other repeatedly as they move through different regions of the cloud. During these collisions, electric charges are transferred between particles, causing some regions of the cloud to become positively charged while others become negatively charged. This separation of charges gradually increases the electric field within the cloud. Establish how collisions between particles inside clouds lead to charge separation.
41.	During stormy weather, large clouds accumulate electric charges that eventually lead to lightning flashes. Identify the source of these electric charges in clouds.
42.	During a science lesson on atmospheric electricity, a teacher explains that thunderclouds often contain large quantities of electric charge. These charges develop because particles such as water droplets, ice crystals, and dust collide frequently as they move within the cloud due to strong air currents. Over time, the charges accumulate in different regions of the cloud and may eventually discharge as lightning during thunderstorms. Determine the process responsible for the formation of electric charges inside clouds.
	Unit 2

1.	In a microelectronics laboratory two equal positive charges are placed very close to each other while engineers attempt to find a point between them where the electric field becomes zero. When a third nearby charge is introduced the field at that point no longer cancels. Determine why the electric field cancellation fails.
2.	A charged particle detector is placed near an isolated positive charge. Engineers observe that whenever a small positive test charge is released near the detector it always moves away from the source charge rather than toward it. Infer the cause of this motion using the concept of electric field direction around a point charge.
3.	A teaching laboratory places three charges around a measurement point forming a triangular arrangement. Students notice that when one charge is removed, the measured electric field direction changes noticeably even though the remaining two charges remain unchanged. This observation leads them to analyze how individual electric field contributions affect the resultant field. Assess how the principle of superposition explains the change in resultant electric field when one of the charges is removed.
4.	During an electrostatic experiment, a charged particle produces an electric field of 12000 N/C at a distance of 1 m from the charge. When the observation point is moved to 2 m away from the same charge, researchers observe that the field intensity decreases significantly. Students are asked to interpret this change based on the mathematical relationship governing electric fields. Determine how the electric field changes with distance and evaluate the new field value at 2 m.
5.	A calibration facility places two point charges along a straight line in order to measure electric field strength at a monitoring point. A charge of +5 microC is located at point A and a charge of +2 microC is located 2 m away from it at point B. A sensor is placed exactly 1 m from each charge along the line between them. Engineers want to determine the combined electric field experienced by the sensor due to both charges. The Coulomb constant is $k = 9 \times 10^9$ N m² per C². Determine the magnitude and direction of the resultant electric field at the sensor location using the principle of superposition.

6.	In a research experiment studying electrostatic interactions, two identical positive charges are fixed at opposite ends of a straight insulating rod while a smaller negative charge is placed somewhere between them. Researchers observe that a tiny charged particle released near the center sometimes moves toward one side instead of staying balanced. The scientists suspect that the combined electric field created by the charges is responsible for this behavior. They note that even though the charges appear symmetrically placed, the presence of the negative charge changes the field pattern within the region. Analyze how the electric field directions produced by the positive and negative charges interact in this region and infer why the test particle experiences a net force instead of remaining in equilibrium.
7.	An electronics laboratory is designing a sensor calibration chamber where several charged metallic probes are placed in fixed positions to create a controlled electric field environment. Three probes carrying charges of +4 microC, −2 microC, and +3 microC are mounted at different points inside the chamber. Engineers notice that the field detected at a test location depends strongly on the combined influence of all three probes rather than on any single charge. During calibration, the sensor output changes whenever one probe is removed or repositioned, indicating that the field at the observation point results from the cumulative interaction of multiple charges acting simultaneously. Using the concept of electric field due to a system of charges, determine how the resultant electric field at the sensor location is influenced by the presence of all probes and analyze how removing one probe would alter the overall field distribution.
8.	While studying electric field diagrams a student suggests that two electric field lines could intersect at a point in space. However the instructor explains that this situation is physically impossible in an electric field representation. Determine why electric field lines cannot intersect at any point.
9.	During a classroom experiment a positive test charge is placed at different points around a negative charge. Students notice that the direction of the electric field arrows drawn in the diagram always point toward the negative charge. Infer why the electric field lines always point toward the negative charge.

10.	A teaching laboratory compares electric field diagrams of two systems. In the first system a single isolated positive charge is present. In the second system a positive charge is placed near a negative charge forming a dipole configuration. Students observe that the field lines in the dipole system begin at the positive charge and terminate at the negative charge rather than spreading uniformly outward. Assess how the electric field line pattern differs between the two systems and what this indicates about the interaction between opposite charges.
11.	In an electric field visualization experiment, students observe that around one charged particle approximately 20 electric field lines are drawn to represent the electric field. When the charge of the particle is doubled while keeping the visualization scale constant, the diagram shows nearly 40 field lines emerging from the charge. Interpret what this change in the number of field lines indicates about the relationship between electric charge and electric field representation.
12.	An electrostatic simulation system is used to study the interaction between two identical positive charges placed close to each other. When the electric field pattern is generated, students observe that the electric field lines originating from each charge curve away from the region between them rather than crossing directly toward the other charge. The space between the charges shows a region where the lines appear to repel each other. Evaluate why electric field lines behave in this manner and determine what this pattern indicates about the interaction between the two positive charges.
13.	In an electrostatic demonstration experiment, two charged metal spheres are placed on an insulated table. One sphere carries a large positive charge while the other carries a smaller positive charge. When researchers map the electric field lines around both spheres, they observe that the number of field lines emerging from the larger charge is significantly higher than those from the smaller charge. Additionally, the lines appear more concentrated close to the surface of the charged spheres. Analyze how the density of electric field lines represents the strength of the electric field and infer why regions closer to the charge show a higher concentration of field lines.

14.	A physics research laboratory is studying electric field visualization using charged metallic spheres. During an experiment, a positively charged sphere is placed in the center of an insulating platform and fine electric field mapping sensors are arranged around it. The sensors record that the electric field appears to spread outward uniformly from the surface of the sphere. When the same experiment is repeated with a negatively charged sphere, the sensors show that the field pattern appears reversed in direction but maintains a similar symmetrical distribution. Researchers use this observation to represent electric fields using field line diagrams. Analyze how the direction and arrangement of electric field lines represent the nature of positive and negative charges and infer why these lines are useful in visualizing electric field behavior.
15.	Researchers studying dielectric materials observe that certain molecules inside the material behave like tiny electric dipoles. When an external electric field is applied the molecules tend to orient themselves along the field direction. Determine why this dipole behavior is important in understanding the electrical properties of materials.
16.	In an electrostatic experiment two equal charges of opposite sign are placed very close together compared to other distances in the system. Scientists refer to this arrangement as an electric dipole. Infer why this pair of charges is classified as a dipole rather than two independent charges.
17.	An electrostatic simulation system generates the electric field pattern around an electric dipole. Students observe that the electric field lines originate near the positive charge and curve toward the negative charge rather than spreading uniformly outward as seen for a single charge. The field intensity appears stronger near the charges and weaker farther away. Assess how this field pattern represents the electric field behavior of an electric dipole.

18.	Two electric dipoles are created in a laboratory. The first dipole has charges of magnitude 2 microC separated by 5 cm, while the second dipole has charges of magnitude 2 microC separated by 10 cm. Students are asked to determine how the difference in separation distance influences the dipole moment. Evaluate which dipole has a larger dipole moment and determine the relative increase in dipole strength.
19.	In an electrostatic experiment two point charges of equal magnitude but opposite sign are placed 4 cm apart forming an electric dipole. Each charge has a magnitude of 3 microC. Researchers want to calculate the electric dipole moment of this system in order to understand the strength of the dipole. Determine the electric dipole moment using the relation $p = q \times d$
20.	In a molecular physics laboratory, researchers analyze polar molecules such as water which behave like electric dipoles due to uneven charge distribution. When these molecules are placed in an external electric field, scientists observe that many of them rotate and align themselves in a particular direction. The molecules initially have random orientations, but after some time the system develops a preferred alignment along the electric field direction. Analyze why dipole molecules tend to align with an external electric field and infer how the dipole moment influences this alignment behavior.
21.	A nanotechnology laboratory is studying the behavior of molecular systems that exhibit electric dipole characteristics. In one experiment, scientists place two equal but opposite charges separated by a small fixed distance on an insulating platform to simulate a dipole. Sensors placed around the setup detect that the electric field distribution in the surrounding space differs significantly from that produced by a single charge. The field pattern shows strong directional variation, and the effect becomes more pronounced when the separation distance between the charges increases while maintaining the same charge magnitude. Analyze how the presence of two equal and opposite charges forms an electric dipole and evaluate how the separation distance influences the dipole's electric field behavior.

22.	Researchers studying electrostatic systems observe that many real objects such as wires plates and insulating materials contain charges spread over their geometry instead of concentrated at one point. Scientists therefore model these systems using continuous charge distributions. Determine why continuous charge distribution models are important for analyzing electric fields in real systems.
23.	During a classroom experiment a long conducting rod is uniformly charged along its entire length. Students are asked to express the distribution of charge along the rod using a suitable physical quantity. Infer why linear charge density is used to describe this distribution.
24.	A simulation program visualizes the electric field produced by a long uniformly charged wire. Students observe that the electric field around the wire forms circular symmetry around the wire and its magnitude decreases as the distance from the wire increases. Researchers explain that the electric field at a point results from the combined contribution of all small charge elements along the wire. Assess why continuous charge distribution requires considering contributions from many small charge elements instead of treating the wire as a single point charge.
25.	A square conducting plate carries a total charge of 8 microC distributed uniformly across its surface. The area of the plate is 2 square meters. Students are asked to determine the surface charge density of the plate in order to analyze the electric field generated by the charged surface. Evaluate the surface charge density using the relation sigma = Q / A.
26.	In a laboratory experiment a thin straight wire carries a total charge of 12 microC uniformly distributed along its length of 3 m. Researchers want to determine the linear charge density of the wire in order to analyze the electric field produced by this continuous charge distribution. Use the relation lambda = Q / L to determine the linear charge density.

27.	A laboratory experiment investigates how electric charge can be distributed in different physical structures. Researchers analyze a long charged conducting wire, a charged metal sheet, and a uniformly charged insulating sphere. They observe that the charge in each case is distributed differently depending on the geometry of the object. To mathematically describe these distributions scientists define quantities that represent charge per unit length, charge per unit area, and charge per unit volume. Analyze how linear surface and volume charge densities represent different types of continuous charge distributions and infer why these quantities are necessary for electric field calculations.
28.	An electrostatic research laboratory studies different ways in which electric charge can be distributed in physical systems. In one experiment, engineers examine three different objects: a thin charged wire, a large charged metal plate, and a uniformly charged solid sphere. Each object contains a large number of charges that are spread continuously rather than concentrated at a single point. Scientists observe that calculating the electric field produced by these objects requires considering the charge distribution over their entire geometry rather than treating them as individual point charges. Analyze how continuous charge distribution differs from discrete point charge systems and evaluate why integration methods are required to determine the electric field produced by such systems.
29.	Researchers studying electrostatic shielding observe that large charged plates produce electric fields that are nearly uniform in the region close to the plate surface. Determine why an infinite plane charge distribution produces a uniform electric field.
30.	During an electrostatics experiment students observe that the electric field produced by an infinitely long charged wire always points radially outward from the wire. Infer why the electric field cannot have a component along the length of the wire.

31.	A computer simulation visualizes the electric field around a very long uniformly charged wire. Students observe that the electric field lines form circular symmetry around the wire and point radially outward. The density of the lines decreases as the distance from the wire increases. Assess what this electric field line pattern indicates about the behavior of the electric field produced by an infinite line charge.
32.	A large metal sheet carries a uniform surface charge density of sigma = 4 × 10^-6 C/m². Researchers want to determine the electric field produced near the sheet. The electric field due to an infinite plane sheet is given by E = sigma / (2 × epsilon0) Calculate the electric field near the sheet.
33.	In an electrostatic experiment a very long charged wire carries a uniform linear charge density of lambda = 5 × 10^-6 C/m. Researchers want to determine the electric field at a point located 0.5 m from the wire. The electric field due to an infinite line charge is given by E = lambda / (2 × pi × epsilon0 × r) where epsilon0 = 8.85 × 10^-12 C²/N·m². Determine the magnitude of the electric field at the observation point.
34.	A research team investigates the electric field produced by a large uniformly charged metal sheet that is very large compared to the observation region. Electric field sensors placed at different distances from the sheet show that the field strength remains nearly constant even when the distance from the sheet changes significantly. The scientists explain that this behavior is different from the electric field produced by a point charge. Analyze why the electric field due to an infinite plane sheet remains constant with distance and infer how the symmetry of the plane affects the field direction.

35.	An electrostatics laboratory studies the electric field produced by very long charged conductors. In one experiment, researchers place a very long straight wire carrying uniform positive charge and measure the electric field at different points around it using sensors. They observe that the electric field magnitude depends only on the radial distance from the wire and remains the same at all points located at the same distance around it. Scientists explain that the symmetry of the charge distribution plays an important role in determining the electric field pattern. Analyze how the symmetry of an infinitely long charged wire influences the electric field distribution and evaluate why the electric field depends only on the perpendicular distance from the wire.
36.	Engineers designing electrostatic shielding devices often use hollow conducting spheres to protect sensitive equipment from external electric fields. Determine why a spherical shell is effective in providing electrostatic shielding.
37.	During a classroom experiment students analyze the electric field inside a hollow metal sphere carrying uniform surface charge. Measurements show that the electric field inside the sphere is zero at all points. Infer why the electric field inside the spherical shell becomes zero.
38.	A computer simulation displays the electric field lines produced by a uniformly charged spherical shell. Students observe that outside the shell the field lines radiate outward from the sphere, while inside the shell no field lines appear in the hollow region. The pattern resembles the field produced by a single point charge located at the center. Assess what this electric field pattern indicates about the behavior of the electric field inside and outside the spherical shell.
39.	A uniformly charged spherical shell carries a total charge distributed evenly over its surface. A sensor is placed at a point inside the hollow cavity at a distance of 0.2 m from the center while the radius of the shell is 0.5 m. Determine the electric field at this internal point and explain the result based on spherical symmetry.

40.	A uniformly charged spherical shell carries a total charge of $Q = 6 \times 10^{-6}$ C and has a radius of R = 0.5 m. Researchers want to calculate the electric field at a point located r = 1 m from the center of the shell. The electric field outside a spherical shell is given by $E = (1 / (4 \times pi \times epsilon0)) \times (Q / r^2)$ where $epsilon0 = 8.85 \times 10^{-12}$ $C^2/N \cdot m^2$. Determine the electric field magnitude at the observation point.
41.	Scientists analyze the electric field produced by a hollow spherical shell carrying uniform positive charge on its surface. They place electric field probes at three regions: inside the hollow cavity, on the surface of the shell, and at points outside the shell. Measurements show that the electric field is zero inside the cavity, reaches a certain value at the surface, and decreases as the distance from the sphere increases. Analyze why the electric field behaves differently in these three regions and infer how the spherical symmetry influences the field distribution.
42.	A research laboratory studies the electric field produced by spherical charge distributions. In one experiment a thin hollow metal sphere is uniformly coated with electric charge over its entire surface. Sensors are placed at different locations both inside and outside the sphere to measure the electric field. The researchers observe that outside the sphere the electric field behaves as if all the charge were concentrated at the center, while inside the hollow region the sensors detect no electric field at all. Analyze how the symmetry of a uniformly charged spherical shell determines the electric field distribution and evaluate why the electric field inside the shell becomes zero.
	Unit 3

1.	A designer working on electromagnetic lifting equipment studies the magnetic field produced by current-carrying coils. The designer notes that stronger magnetic fields improve lifting capacity and considers increasing the coil current in the electromagnet. Determine how current variation affects magnetic field generation around the coil. Infer how this principle supports the design of electromagnetic lifting systems.
2.	A magnetic core inside a transformer experiences a magnetic flux of 0.02 Weber passing through a cross-sectional area of 0.01 m^2. Engineers need to determine the magnetic flux density inside the core to evaluate the performance of the transformer core material. Formula Hint: $B = \Phi / A$ Determine the magnetic flux density in the transformer core.
3.	A student observes magnetic field lines around a bar magnet placed near iron filings on a laboratory table. The filings arrange themselves along curved patterns extending from the north pole to the south pole of the magnet. The student notes that these patterns visually represent the direction and strength of the magnetic field around the magnet. Determine how magnetic field lines represent the direction of magnetic forces around a magnet. Evaluate how the density of field lines indicates the strength of the magnetic field.
4.	An engineer testing an electromagnet observes that increasing current in the coil significantly strengthens the magnetic field produced by the device. The electromagnet is connected to a DC supply and the current flowing through the coil increases from 2 A to 6 A while the coil geometry remains unchanged. The engineer investigates the relationship between electric current and magnetic field generation around the conductor. Formula Hint: $B \propto I$ Analyze why increasing current increases magnetic field strength in the electromagnet. Infer how current variation influences electromagnetic device performance.
5.	A power systems laboratory investigates the magnetic flux produced in an electromagnetic coil used in transformers. The coil carries a magnetic field across a cross-sectional area of 0.02 m^2 and the magnetic flux density inside the core is measured as 0.5 Tesla. Engineers need to determine the total magnetic flux passing through the core area. This experiment helps students understand how magnetic flux is related to flux density and cross-sectional area in electromagnetic devices such as transformers and inductors. Formula Hint: $\Phi = B \times A$ Determine the magnetic flux produced in the electromagnetic core. Interpret how increasing the cross-sectional area of the magnetic core would influence the magnetic flux.
6.	An electric motor design team analyzes how magnetic fields influence the motion of conductors carrying electric current inside motor windings. A straight conductor placed inside a magnetic field carries a current of 5 A, and the magnetic flux density in the motor air gap is 0.4 Tesla. The length of the conductor inside the magnetic field region is 0.3 m. Engineers must determine the force acting on the conductor because this force produces the mechanical rotation required for motor operation. The interaction between the magnetic field and the current-carrying conductor follows the electromagnetic force law used in motor design. Formula Hint: $F = B \times I \times L$ Determine the electromagnetic force acting on the conductor inside the magnetic field. Analyze how increasing magnetic field strength would influence the mechanical output of the motor.

7.	An electrical engineering laboratory is studying the magnetic field generated by a long straight current-carrying conductor used in electromagnetic devices. A copper wire carries a steady current of 8 A, and researchers measure the magnetic field intensity at a point 0.05 m away from the wire. The engineers know that the magnetic field around a long straight conductor depends on the current flowing through the wire and the radial distance from the conductor. The relationship between magnetic field strength and current is expressed using the Biot-Savart derived relation for a straight conductor. Students are asked to compute the magnetic field intensity at the given distance to understand how current influences magnetic field formation around conductors used in motors and generators. Formula Hint: $B = (\mu_0 \times I) / (2\pi r)$ $\mu_0 = 4\pi \times 10^{-7}$ T·m/A Determine the magnetic field intensity produced around the conductor at the specified distance. Analyze how increasing the current through the conductor would influence the magnetic field magnitude in this experimental setup.
8.	An electrical engineer designing a busbar system must ensure that magnetic interference around high-current conductors is minimized in nearby electronic equipment. Formula Hint: $B \propto I$ $B \propto 1/r$ Determine how increasing distance from the conductor influences magnetic field exposure.
9.	An electrical engineering student writes the following code to compute the magnetic field near a current-carrying wire: $B = (\mu_0 \times I) / (2 \times r)$ However, the computed value is incorrect because a key mathematical term is missing. Formula Hint: Correct Formula: $B = (\mu_0 \times I) / (2 \times \pi \times r)$ Identify the error in the program and determine the correction required
10.	During a laboratory demonstration, a current-carrying wire is placed vertically and iron filings are sprinkled around it while current flows through the conductor. The filings arrange themselves into circular patterns centered on the wire. Determine how the right-hand thumb rule explains the direction of magnetic field lines around the conductor.
11.	A student performs an experiment to measure the magnetic field around a straight conductor but notices that the measured value is lower than expected when the probe is moved further away from the wire. The student analyzes how the distance from the conductor influences the magnetic field intensity. Formula Hint: $B \propto 1 / r$ Analyze why the magnetic field decreases as the distance from the conductor increases.

12.	A laboratory experiment measures the magnetic field produced around a long straight copper conductor carrying a steady current of 12 A. The measurement probe is placed at a distance of 0.03 m from the wire. Engineers want to determine the magnetic field intensity at this point using the theoretical formula derived from Ampere's Circuital Law. Formula Hint: B = (μ0 × I) / (2 × π × r) μ0 = 4 × π × 10^-7 Determine the magnetic field intensity at the specified point and analyze how reducing the distance from the conductor would affect the magnetic field strength.
13.	An engineering student writes a program to calculate the magnetic field around a current-carrying wire but forgets to complete some statements. The program should calculate the magnetic field intensity at a point located a certain distance from the wire using Ampere's law. The program structure is partially written as shown below. Program Snippet: Input: Current I, Distance r Constant μ0 = 4 × π × 10^-7 Step1: _____ Step2: B = (μ0 × I) / (2 × π × r) Step3: _____ Formula Hint: B = (μ0 × I) / (2 × π × r) Determine the missing steps in the program so that the system correctly calculates and displays the magnetic field.
14.	An electrical engineering student is developing a small computational tool to estimate the magnetic field produced by an infinitely long straight current-carrying conductor used in power transmission experiments. The program receives two inputs: the current flowing through the wire and the distance of the observation point from the conductor. The student must design a logical sequence that calculates the magnetic field using the theoretical expression derived from Ampere's Circuital Law. The program should compute the magnetic field value and display it in Tesla so that engineers can quickly estimate magnetic field strength near electrical conductors. Formula Hint: B = (μ0 × I) / (2 × π × r) μ0 = 4 × π × 10^-7 Formulate a pseudocode procedure that accepts current and distance as input and computes the magnetic field intensity using the above expression. Verify how the algorithm ensures correct magnetic field calculation.
15.	Magnetic field inside a solenoid depends on turns per unit length and current. Formula Hint: B = μ0 × n × I Determine how increasing the number of turns affects the magnetic field.
16.	A solenoid has 800 turns, length 0.40 m, and carries 2 A current. Formula Hint: B = μ0 × n × I n = N / L Determine the magnetic field strength.
17.	Toroidal inductors are widely used in electrical circuits. Determine why toroidal coils are preferred in many electrical devices.

18.	A long solenoid connected to a DC supply produces a strong magnetic field inside the coil. Determine why the magnetic field inside the solenoid remains nearly uniform.
19.	A toroidal coil used in an inductor contains 400 turns, carries 3 A current, and has an average radius of 0.05 m. Formula Hint: B = (μ0 × N × I) / (2πr) μ0 = 4π × 10^-7 Determine the magnetic field inside the toroid.
20.	Two solenoids operate with the same current 2 A. Solenoid X has 1500 turns/m, while Solenoid Y has 2500 turns/m. Formula Hint: B = μ0 × n × I Determine the magnetic fields of both solenoids and identify which produces the stronger magnetic field.
21.	A solenoid with 1000 turns is wound uniformly over a length of 0.50 m. The coil carries 1.8 A current. Engineers need to calculate the magnetic field generated inside the solenoid. Formula Hint: B = μ0 × n × I n = N / L μ0 = 4π × 10^-7 Determine the magnetic field strength inside the solenoid.
22.	In atomic magnetism, the magnetic dipole moment is directly proportional to the current and the area of the orbit. Formula Hint: μ = I × A Determine how increasing the radius of the electron orbit affects the magnetic dipole moment.
23.	A circular electron orbit has radius 4×10^{-11} m and carries an effective current of 2×10^{-3} A. Formula Hint: Magnetic Dipole Moment μ = I × A Area A = πr^2 Determine the magnetic dipole moment of the orbiting electron.
24.	An atomic model shows an electron revolving in a circular orbit around the nucleus. The direction of the resulting magnetic dipole moment depends on the direction of the electron's motion. Scientists use a rule similar to the right-hand rule to determine the direction of the magnetic dipole moment generated by the moving charge. Determine how the direction of the magnetic dipole moment is related to the electron's orbital motion.
25.	A student observes that atoms containing unpaired electrons often exhibit magnetic behavior, while atoms with paired electrons usually show weak or no magnetism. This observation relates to the way magnetic dipole moments of electrons combine within an atom. Determine why paired electrons cancel magnetic effects while unpaired electrons contribute to atomic magnetism.

26.	In an advanced electromagnetism course, students analyze the relationship between angular momentum and magnetic dipole moment of a revolving electron. When an electron moves in a circular orbit, it possesses angular momentum and also produces a magnetic moment due to its motion. Physicists have shown that the magnetic dipole moment of the electron is proportional to its angular momentum. This relationship explains the origin of atomic magnetism and is fundamental to understanding the magnetic properties of atoms and molecules. Formula Hint: $\mu = (e / 2m) \times L$ Where: μ = magnetic dipole moment L = angular momentum e = electron charge m = electron mass Determine how angular momentum influences the magnetic dipole moment of an electron and interpret the significance of this relationship in atomic magnetism.
27.	A materials science engineer studies the magnetic behavior of atoms in a ferromagnetic material. The engineer explains that each electron revolving around the nucleus produces a magnetic dipole moment similar to a tiny current loop. When many electrons in a material align their magnetic dipole moments in the same direction, the material exhibits strong magnetism. However, if the dipole moments are randomly oriented, the overall magnetic effect becomes negligible. Understanding this concept helps engineers design materials used in electrical machines and magnetic storage devices. Formula Hint: Magnetic Dipole Moment $\mu = I \times A$ Direction determined by Right Hand Rule Analyze how the motion of electrons produces magnetic dipole moments and determine why alignment of atomic dipoles leads to strong magnetization in materials.
28.	In an atomic physics laboratory, researchers analyze how a revolving electron generates a magnetic dipole moment due to its circular motion around the nucleus. Consider an electron moving in a circular orbit of radius 5×10^{-11} m with a velocity of 2×10^{6} m/s. The electron behaves like a small current loop because a moving charge constitutes an electric current. Engineers studying electromagnetic properties of atoms must determine the magnetic dipole moment produced by this orbiting electron. Formula Hint: Magnetic Dipole Moment (μ) = I × A Current I = q / T Time period $T = 2\pi r / v$ Area $A = \pi r^2$ Charge of electron q = 1.6 × 10^-19 C Determine the magnetic dipole moment of the revolving electron and analyze how increasing the electron velocity would influence the magnetic dipole moment.
29.	Magnetic field of a dipole depends on the cube of the distance from the magnet. Formula Hint: $B \propto 1 / r^3$ Determine how doubling the distance affects the magnetic field strength.

30.	A magnetic dipole has magnetic moment 0.3 A·m². A point lies 0.20 m away from the magnet along the axial line. Formula Hint: B_axial = (μ0 / 4π) × (2M / r³) μ0 / 4π = 10^-7 Determine the magnetic field intensity.
31.	Magnetic dipole fields are used in magnetic sensors and compasses to detect the Earth's magnetic field. Determine why understanding axial and equatorial magnetic fields is important in designing magnetic sensing devices.
32.	During a laboratory experiment, iron filings are sprinkled around a bar magnet placed on a sheet of paper. The filings form curved patterns emerging from one pole of the magnet and entering the other pole. Determine how magnetic field lines represent the magnetic dipole field pattern around a bar magnet.
33.	A bar magnet has magnetic moment 0.6 A·m². A sensor measures the magnetic field at a distance 0.05 m from the magnet along both axial and equatorial directions. Engineers want to compare the field strength in both directions to understand dipole field distribution used in magnetic sensing systems. Formula Hint: Axial field: B_axial = (μ0 / 4π) × (2M / r³) Equatorial field: B_equatorial = (μ0 / 4π) × (M / r³) μ0 / 4π = 10^-7 Determine both magnetic fields and analyze which one is stronger.
34.	A magnetic dipole with magnetic moment 0.4 A·m² is placed in a laboratory setup for studying magnetic field distribution. A magnetic field sensor is placed at a point 0.08 m from the center of the magnet along the equatorial line, which is perpendicular to the magnet's axis. Engineers need to determine the magnetic field intensity at this location to compare axial and equatorial magnetic field strengths. Formula Hint: Magnetic field on equatorial line: B_equatorial = (μ0 / 4π) × (M / r³) μ0 / 4π = 10^-7 Determine the magnetic field intensity along the equatorial line and interpret how it compares with the axial field.
35.	A small bar magnet used in a laboratory experiment behaves like a magnetic dipole. The magnetic moment of the magnet is 0.5 A·m², and a measurement probe is placed 0.10 m away from the center of the magnet along its axial line. Engineers studying magnetic field distribution want to determine the magnetic field intensity at this point. Understanding the axial magnetic field helps in designing magnetic sensors and analyzing magnetic material behavior. Formula Hint: Magnetic field on axial line of dipole: B_axial = (μ0 / 4π) × (2M / r³) μ0 / 4π = 10^-7 Determine the magnetic field intensity along the axial line and analyze how increasing the distance affects the field strength.

36.	Faraday's law states induced EMF depends on rate of flux change. Formula Hint: $E \propto d\Phi / dt$ Determine effect of halving the flux change rate.
37.	A coil with 60 turns experiences flux change 0.012 Weber in 0.006 seconds. Formula Hint: $E = - N \times (\Delta\Phi / \Delta t)$ Determine induced EMF.
38.	Electric generators convert mechanical energy into electrical energy using rotating coils in magnetic fields. Determine why electromagnetic induction enables energy generation in generators.
39.	When a magnet moves toward a conducting coil, a current is induced in the coil. Determine why the induced current direction opposes the magnet's motion.
40.	Two coils experience identical magnetic flux change rate of 0.02 Wb/s. Coil X has 150 turns, while Coil Y has 300 turns. Formula Hint: $E = - N \times (d\Phi / dt)$ Determine the induced EMF in both coils.
41.	A coil of 250 turns experiences magnetic flux changing at 0.015 Weber per second. Formula Hint: $E = - N \times (d\Phi / dt)$ Determine the induced EMF.
42.	A coil with 120 turns experiences a decrease in magnetic flux from 0.03 Weber to 0.01 Weber in 0.02 seconds due to movement of a magnet away from the coil. Formula Hint: $E = - N \times (\Delta\Phi / \Delta t)$ Determine the induced EMF in the coil.
	Unit 4
1.	A charged particle moves perpendicular to a magnetic field. Formula Hint $F = q \times v \times B$ Determine how doubling the magnetic field affects the magnetic force.
2.	A charge 4×10^{-6} C is placed in an electric field of 2500 N/C. Formula Hint $F = q \times E$ Determine the electric force acting on the charge.
3.	A proton moves in a region where both electric and magnetic fields are present. Engineers analyze how each field affects the motion of the particle. Determine why electric fields can change the speed of a charged particle while magnetic fields mainly change the direction of motion.

4.	A charged particle enters a magnetic field with its velocity parallel to the direction of the magnetic field. Engineers studying electromagnetic forces analyze the resulting magnetic force acting on the particle. Formula Hint $F = q \times v \times B \times \sin\theta$ Determine the magnetic force acting on the particle and explain the result.
5.	A charged particle moves through a region where both electric and magnetic fields are present simultaneously. The particle experiences a combined electromagnetic force known as the Lorentz force. Engineers analyzing the motion of charged particles in devices such as cathode ray tubes and particle accelerators must determine how the electric field and magnetic field together influence the motion of the particle. Formula Hint Lorentz Force $F = q\,(E + v \times B)$ Explain how the Lorentz force determines the motion of a charged particle when both electric and magnetic fields act simultaneously.
6.	An electron with charge 1.6×10^{-19} C moves with velocity 2.5×10^{6} m/s perpendicular to a uniform magnetic field of 0.015 Tesla. The motion of the electron in the magnetic field produces a magnetic force which changes the direction of motion of the electron. Engineers studying particle motion in electromagnetic systems must determine the magnitude of the magnetic force acting on the electron. Formula Hint Magnetic Force $F = q \times v \times B$ Determine the magnetic force acting on the electron and interpret how increasing the velocity of the particle affects the magnetic force.
7.	A charged particle with charge 3×10^{-6} C is placed in a uniform electric field of strength 4×10^{3} N/C. The electric field exerts a force on the charge causing it to accelerate in the direction of the field. Engineers studying charged particle behavior in electric fields must determine the magnitude of the electric force acting on the particle and understand how the electric field strength influences this force. Formula Hint Electric Force $F = q \times E$ Determine the electric force acting on the charge and analyze how increasing the electric field strength would influence the magnitude of the force.
8.	A charge moves in a uniform electric field. Formula Hint $F \propto E$ Determine the effect of tripling the electric field strength.
9.	A charge 3×10^{-6} C is placed in an electric field 2000 N/C. Formula Hint $F = q \times E$ Determine the electric force.
10.	Electrons moving in cathode ray tubes pass through electric fields and experience deflection. Determine why electric fields can change the motion of electrons.

11.	A negatively charged particle is placed in a uniform electric field between two plates. Determine the direction of force acting on the particle.
12.	A charged particle with mass 4×10^{-6} kg and charge 3×10^{-6} C moves inside a uniform electric field of 3×10^{3} N/C. Formula Hint $F = q \times E$ $F = m \times a$ Determine the acceleration of the particle due to the electric field.
13.	An electron moves inside a region where a uniform electric field of 3×10^{4} N/C exists between two plates. Engineers analyzing particle accelerators must determine the electric force acting on the electron. Charge of electron = 1.6×10^{-19} C Formula Hint $F = q \times E$ Determine the magnitude of the electric force acting on the electron.
14.	A particle with charge 4×10^{-6} C is placed in a uniform electric field of 5×10^{3} N/C between two conducting plates. Engineers analyzing electrostatic forces must determine the magnitude of the force acting on the particle and understand how the electric field controls particle motion. Formula Hint $F = q \times E$ Determine the electric force acting on the charge and analyze how increasing the charge would affect the force.
15.	A charged particle moves in a magnetic field perpendicular to its velocity. Formula Hint $F \propto B$ Determine how doubling the magnetic field affects the magnetic force.
16.	A charge 3×10^{-6} C moves with velocity 2×10^{5} m/s in a magnetic field of 0.01 Tesla. Formula Hint $F = q \times v \times B$ Determine the magnetic force acting on the charge.
17.	A beam of charged particles enters a uniform magnetic field and begins to move in circular paths. Engineers use this principle in instruments such as mass spectrometers and cyclotrons. Determine why charged particles follow circular paths in a magnetic field.
18.	A positively charged particle moves through a magnetic field perpendicular to its velocity. Engineers studying magnetic field effects must determine the direction of the magnetic force acting on the particle. Determine how the right-hand rule is used to identify the direction of the magnetic force.

19.	A charged particle enters a magnetic field at an angle to the field direction. Because the velocity has components both perpendicular and parallel to the magnetic field, the particle follows a helical trajectory in the magnetic field. Engineers analyzing magnetic confinement systems must understand the relationship between particle motion and magnetic force. Formula Hint $F = q \times v \times B \times \sin\theta$ Explain how the magnetic force influences the motion of the particle and why the particle follows a helical path.
20.	A proton with charge 1.6×10^{-19} C enters a uniform magnetic field of 0.03 Tesla with velocity 3×10^{6} m/s perpendicular to the field direction. The magnetic field exerts a force that causes the proton to move in a circular path. Engineers studying cyclotron devices must determine the magnetic force acting on the proton. Formula Hint Magnetic Force $F = q \times v \times B$ Determine the magnetic force acting on the proton and interpret how increasing the magnetic field strength affects the force.
21.	A charged particle with charge 2×10^{-6} C moves with velocity 4×10^{5} m/s perpendicular to a uniform magnetic field of 0.02 Tesla. The interaction between the moving charge and the magnetic field produces a magnetic force that changes the direction of motion of the particle. Engineers analyzing the behavior of charged particles in magnetic confinement systems must determine the magnitude of the magnetic force acting on the particle. Formula Hint Magnetic Force $F = q \times v \times B$ Determine the magnetic force acting on the particle and analyze how increasing the velocity of the particle affects the magnetic force.
22.	A velocity selector uses crossed electric and magnetic fields. Formula Hint $v \propto E$ $v \propto 1/B$ Determine how doubling the magnetic field affects the selected velocity.
23.	A velocity selector has electric field 2000 N/C and magnetic field 0.02 Tesla. Formula Hint $v = E / B$ Determine the velocity of particles that pass through undeflected.
24.	Scientists use crossed electric and magnetic fields in instruments such as mass spectrometers to control particle motion. Determine why velocity selectors are important in such instruments.
25.	In a velocity selector experiment, charged particles enter a region where electric and magnetic fields are perpendicular to each other. Some particles pass straight through the device without deflection while others deviate from the path. Determine why only particles with a specific velocity pass through the selector without deflection.

26.	A charged particle moves through a region where electric and magnetic fields act simultaneously but are not balanced. As a result, the particle experiences both electric and magnetic forces which alter its trajectory. Engineers studying particle motion in electromagnetic devices must analyze how the combined forces influence the path of the particle. Formula Hint Lorentz Force $F = q(E + v \times B)$ Explain how the Lorentz force determines the trajectory of a charged particle when electric and magnetic fields act together.
27.	A charged particle moves through crossed electric and magnetic fields where the electric field strength is 3×10^3 N/C and the magnetic field strength is 0.01 Tesla. Scientists using a particle velocity selector must determine the velocity of particles that remain undeflected when both fields act simultaneously. Formula Hint $v = E / B$ Determine the velocity required for the particle to pass through the velocity selector without deflection and analyze how increasing the electric field influences the selected velocity.
28.	In a laboratory experiment, a beam of charged particles enters a region where a uniform electric field of 5×10^3 N/C and a magnetic field of 0.02 Tesla act perpendicular to each other. The particle beam moves perpendicular to both fields. Engineers studying velocity selector devices must determine the velocity at which the electric and magnetic forces balance so that the particle travels without deflection through the fields. Formula Hint Electric Force $F_E = q \times E$ Magnetic Force $F_B = q \times v \times B$ For balanced motion $qE = qvB$ $v = E / B$ Determine the velocity of the particle that allows it to pass through the crossed electric and magnetic fields without deflection and interpret how this principle is used in velocity selector devices.
29.	A current carrying conductor is placed in a magnetic field. Formula Hint $F \propto I$ Determine how doubling the current affects the magnetic force.
30.	A conductor of length 0.40 m carries a current of 3 A in a magnetic field of 0.10 Tesla. Formula Hint $F = B \times I \times L$ Determine the force acting on the conductor.
31.	In electric motors, a current carrying conductor placed in a magnetic field produces mechanical motion due to magnetic forces acting on the conductor. Determine why this principle is essential for the operation of electric motors.

32.	A current carrying conductor is placed in a magnetic field and experiences a force whose direction depends on the orientation of the current and the magnetic field. Determine how Fleming's left-hand rule is used to identify the direction of force acting on the conductor.
33.	A straight conductor carrying current is placed in a magnetic field and experiences a force due to the interaction between moving charges and the magnetic field. This phenomenon is widely used in electric motors where electrical energy is converted into mechanical motion. Formula Hint $F = B \times I \times L$ Explain how the interaction between current and magnetic field produces a force on the conductor and how this principle enables the operation of electric motors.
34.	A conductor of length 0.30 m carrying a current of 5 A is placed in a uniform magnetic field of 0.40 Tesla. The conductor is positioned perpendicular to the magnetic field. Engineers studying electromagnetic devices must determine the force acting on the conductor and understand how magnetic fields produce mechanical motion in electric motors. Formula Hint $F = B \times I \times L$ Determine the force acting on the conductor and analyze how increasing the magnetic field strength affects the force.
35.	A straight conductor of length 0.50 m carries a current of 4 A and is placed perpendicular to a uniform magnetic field of 0.20 Tesla. When current flows through the conductor, it experiences a magnetic force due to the interaction between the magnetic field and the moving charges inside the conductor. Engineers analyzing the operation of electric motors must determine the magnitude of the force acting on the conductor. Formula Hint Force on current carrying conductor $F = B \times I \times L$ Determine the magnetic force acting on the conductor and interpret how increasing the current affects the force acting on the conductor.
36.	Two parallel wires carry currents in opposite directions. Determine the nature of the force between them.
37.	Two conductors carry currents 3 A and 2 A and are separated by 0.04 m. A 0.5 m segment is considered. Formula Hint $F = (\mu0 \times I1 \times I2 \times L) / (2\pi \times d)$ $\mu0 = 4\pi \times 10^{-7}$ Determine the force between the conductors.
38.	In electrical transmission lines, multiple conductors carrying current are placed close to each other. Engineers must understand the magnetic forces that arise between the conductors. Determine why these forces must be considered when designing power transmission systems.
39.	Two parallel wires carry electric currents in the same direction and are placed close to each other. Determine the nature of the force between the wires and explain why this occurs.

40.	In an experiment involving two long parallel conductors carrying electric currents, each conductor produces a magnetic field around it. This magnetic field interacts with the current in the other conductor and produces a force between them. Engineers must analyze the physical mechanism responsible for this interaction. Formula Hint Magnetic interaction principle $F \propto I1 \times I2$ $F \propto 1 / d$ Explain how the magnetic field produced by one conductor exerts a force on the other conductor and why the direction of current determines whether the wires attract or repel.
41.	Two parallel wires carry currents of 5 A and 4 A respectively and are separated by 0.02 m. Engineers analyzing electrical transmission systems study the force acting on a 1 m length of the wires caused by the magnetic fields produced by each conductor. Formula Hint $F = (\mu0 \times I1 \times I2 \times L) / (2\pi \times d)$ $\mu0 = 4\pi \times 10^{-7}$ Determine the force between the two conductors and explain how the force changes if the distance between them increases.
42.	Two long parallel conductors are placed 0.05 m apart in a laboratory setup used to study electromagnetic interaction between current carrying wires. The first conductor carries a current of 8 A, while the second conductor carries 6 A. A segment of length 0.50 m of the conductors is considered for analysis. Engineers must determine the magnitude of the force between the conductors in order to understand the magnetic interaction between current carrying wires used in power transmission systems. Formula Hint Force between parallel conductors $F = (\mu0 \times I1 \times I2 \times L) / (2\pi \times d)$ $\mu0 = 4\pi \times 10^{-7}$ Determine the force between the conductors and interpret whether the conductors attract or repel if both currents flow in the same direction.
	Unit 5
1.	During an experiment on wireless communication, a transmitter generates electromagnetic waves that propagate through space and reach a receiver placed several meters away. Even though there is no physical conductor connecting the transmitter and receiver, the receiver successfully detects the signal energy carried by the wave. Determine the phenomenon responsible for this energy transfer.
2.	Solar energy travels from the Sun to Earth through empty space. Identify the mechanism responsible for this energy transfer.

3.	In a laboratory demonstration, a transmitter produces electromagnetic waves that propagate through space and reach a detector placed at some distance. The detector records the arrival of energy even though there is no direct electrical connection between the transmitter and the detector. Establish how energy is transferred between the transmitter and detector.
4.	During a microwave transmission experiment, energy generated by a transmitter travels through space and is detected by a receiver placed several meters away. Relate this observation to energy transfer in electromagnetic fields.
5.	Scientists studying solar radiation observe that energy from the Sun reaches Earth even though there is no material medium between them. The energy travels through empty space as electromagnetic radiation consisting of oscillating electric and magnetic fields. This radiation carries energy across millions of kilometers before interacting with matter on Earth. Assess how electromagnetic fields enable energy transfer across empty space.
6.	In modern communication systems such as radio broadcasting, energy produced by a transmitter travels through space before reaching receivers located several kilometers away. During this process, electromagnetic waves carry energy through oscillating electric and magnetic fields. Engineers monitor the strength of these fields at different distances and observe that energy is continuously transported by the electromagnetic field surrounding the wave. Evaluate the role of electromagnetic fields in transporting energy in communication systems.

7.	In a wireless power transmission experiment, engineers design a system where energy is transferred from a transmitting antenna to a receiving antenna placed some distance away. The transmitter generates oscillating electric and magnetic fields that propagate through space. Sensors placed between the two antennas detect the presence of both electric and magnetic components, indicating that energy is being carried by the electromagnetic field itself rather than by physical conductors. The engineers analyze how energy moves through the surrounding space from the source to the receiver. Analyze how electromagnetic fields enable the transfer of energy from a source to a receiving system.
8.	During a classroom experiment, a coil connected to a battery produces a magnetic field around it. When the battery is disconnected, the magnetic field disappears and the stored energy returns to the circuit. Determine the phenomenon illustrated in this experiment.
9.	An inductor connected to a circuit stores energy while current flows through it. Identify where this energy is stored.
10.	In a laboratory demonstration, an inductor connected to a power supply creates a magnetic field around its coil. Students observe that when the supply is suddenly turned off, the magnetic field collapses and releases the stored energy back into the circuit. Establish the relationship between magnetic fields and energy storage.
11.	A coil connected to a power source develops a magnetic field when current flows through it. Relate this observation to energy storage in magnetic fields.

12.	During an experiment with an electromagnet, a student observes that when current flows through a coil wrapped around an iron core, a strong magnetic field forms around the coil. The field exists only while current flows and disappears when the current is switched off. The student learns that the magnetic field contains stored energy that depends on the current in the coil. Assess how magnetic fields act as a medium for storing energy.
13.	In power electronic systems, inductors are commonly used to store energy temporarily in magnetic fields. Engineers design inductors so that when current passes through them, energy is stored in the magnetic field created around the coil. This stored energy can later be released when the current changes, helping regulate electrical circuits. Evaluate the role of magnetic fields in storing electrical energy in inductors.
14.	In an electrical engineering laboratory, students connect a coil to a power supply and observe the behavior of the magnetic field produced around the coil. As current flows through the coil, a magnetic field develops around it and energy becomes stored within that magnetic field. When the power supply is switched off, the magnetic field collapses and releases the stored energy back into the circuit. Analyze how energy is stored in a magnetic field produced by an electric current.
15.	During a classroom demonstration, a teacher moves a magnet toward and away from a coil connected to a galvanometer. Students observe that the galvanometer deflects whenever the magnet is moving but returns to zero when the magnet remains stationary. Determine the concept responsible for this observation.
16.	A rotating coil inside a magnetic field produces electric current. Identify the phenomenon responsible for this effect.
17.	During an experiment, a coil connected to a measuring device produces a current only when a nearby magnet is moving relative to it. Establish why current appears only during motion.

18.	A magnet is moved near a coil connected to a galvanometer, causing the galvanometer to deflect. Relate this observation to electromagnetic induction.
19.	In transformer systems used in power distribution, alternating current flowing through one coil produces a changing magnetic field. This varying magnetic field interacts with a nearby coil and induces an electromotive force in it. As a result, electric energy is transferred between the two coils without direct electrical contact. Assess how changing magnetic fields produce induced emf in nearby conductors.
20.	In electrical power generators, mechanical rotation of coils within magnetic fields produces electric current that is supplied to homes and industries. As the coil rotates, the magnetic field linking the coil continuously changes. This changing magnetic field produces an induced electromotive force that drives current through the external circuit. Evaluate how electromagnetic induction enables electricity generation in generators.
21.	In a laboratory experiment, a student moves a bar magnet toward a coil connected to a sensitive galvanometer. As the magnet approaches the coil, the galvanometer shows a deflection indicating the presence of an electric current. When the magnet is pulled away, the deflection occurs in the opposite direction. The student observes that the current appears only when the magnet is moving relative to the coil. Analyze how electromagnetic induction leads to the production of induced electromotive force in the coil.
22.	In a classroom experiment, a teacher moves a strong magnet close to a thick aluminum plate. Students notice that the plate slightly resists the motion of the magnet. The teacher explains that this happens because changing magnetic fields induce circular electric currents inside the conductor. Determine the phenomenon responsible for this effect.
23.	A metallic conductor placed in a changing magnetic field develops circulating electric currents within it. Identify these currents.

24.	During a laboratory demonstration, a copper disc rotates between the poles of a magnet. Students observe that the disc gradually slows down even though no mechanical friction is applied. The instructor explains that the magnetic field induces circulating currents inside the metal disc which oppose its motion. Establish how eddy currents influence the motion of conductors in magnetic fields.
25.	A metal plate moving between the poles of a magnet experiences resistance to its motion. Relate this effect to the formation of eddy currents.
26.	In magnetic braking systems used in trains and amusement park rides, a strong magnet is placed near a moving metal disc. As the disc rotates through the magnetic field, currents are induced within the metal. These currents interact with the magnetic field and produce a force that slows down the motion of the disc without direct physical contact. Assess how eddy currents are utilized in practical braking systems.
27.	Engineers designing high-speed electric motors notice that solid metallic cores inside rotating magnetic systems produce unwanted heating. Investigations reveal that rapidly changing magnetic fields inside the motor induce circulating currents within the metal core. These currents dissipate electrical energy as heat, reducing the efficiency of the machine. To reduce this effect, engineers use laminated cores instead of solid metal blocks. Evaluate how eddy currents influence electrical machines and how their effects can be reduced.
28.	In a physics laboratory, a metal plate is placed between the poles of a strong magnet. When the plate is moved rapidly within the magnetic field, scientists observe that circular currents are produced inside the conducting plate. These currents create their own magnetic fields which oppose the motion of the plate, making it slightly difficult to move. Engineers analyze this phenomenon to understand how electric currents can be generated inside conductors simply due to changes in magnetic fields. Analyze the formation of eddy currents in a conductor placed in a changing magnetic field.

29.	During a transformer demonstration, a teacher connects one coil to an alternating current source while another nearby coil is connected to a measuring device. Students observe that the second coil shows a voltage even though it is not electrically connected to the first coil. Determine the principle responsible for this observation.
30.	A changing current in one coil produces emf in another nearby coil. Identify the phenomenon responsible.
31.	In a laboratory setup, a coil connected to a battery produces a magnetic field. When the current through the coil changes suddenly, an electromotive force appears within the same coil and opposes the change in current. Establish the phenomenon responsible for this behavior.
32.	Two coils placed close to each other allow energy transfer through changing magnetic fields. Relate this observation to mutual inductance.
33.	During a circuit experiment, a student observes that when the current through a coil is suddenly increased, the coil briefly resists the change in current. This happens because the magnetic field associated with the coil changes and produces an induced electromotive force that opposes the change. Assess how self inductance affects current changes in electrical circuits.
34.	In transformer systems used in power distribution, two coils are placed close to each other. When alternating current flows through the first coil, a changing magnetic field is produced around it. This changing magnetic field interacts with the second coil and induces an electromotive force in it even though the two coils are not directly connected by wires. Evaluate the concept of mutual inductance between two coils.

35.	In an electrical laboratory, a coil connected to a power supply produces a magnetic field around it when current flows. When the current in the coil suddenly changes, the magnetic field around it also changes. This change in the magnetic field induces an electromotive force within the same coil that opposes the change in current. Engineers recognize this behavior as an important property of coils used in electrical circuits. Analyze the concept of self inductance in an electrical coil.
36.	During a laboratory experiment, a focused laser beam is directed toward microscopic particles suspended in a transparent medium. Students observe that the particles slightly move or rotate when exposed to the beam. The instructor explains that this occurs because electromagnetic radiation can transfer both linear and rotational effects to matter. Determine the property of electromagnetic fields responsible for this behavior.
37.	Electromagnetic radiation exerts a small pressure when it strikes a surface. Identify the property responsible for this effect.
38.	In an optical experiment, a laser beam directed at tiny particles causes them to move slightly in the direction of the beam. Establish the reason for this motion.
39.	Sunlight striking a reflective surface produces a small measurable pressure. Relate this effect to the momentum carried by electromagnetic waves.
40.	Researchers studying advanced optical systems observe that certain types of electromagnetic waves can cause tiny particles to rotate when exposed to circularly polarized light. The rotating motion indicates that the electromagnetic field carries angular momentum which can be transferred to matter during interaction. Assess how electromagnetic fields can possess angular momentum.
41.	In space research experiments, scientists use powerful laser beams to manipulate tiny particles. When the laser beam strikes a microscopic particle, it pushes the particle slightly in the direction of the beam. This happens because the electromagnetic field of the laser carries momentum that is transferred to the particle upon interaction. Evaluate how electromagnetic radiation transfers momentum to matter.

42.	Scientists studying solar radiation observe that sunlight not only carries energy but also exerts a small pressure on surfaces it strikes. Sensitive instruments detect that when electromagnetic waves from the Sun hit a reflective surface, a tiny force is produced. This force indicates that electromagnetic radiation carries momentum even though it consists only of electric and magnetic fields traveling through space. Analyze how electromagnetic fields can carry linear momentum.